Suggested Design Project REPORT

Contents to be included

Deadline: 7 June 2026 23:59 Geneva time

Section in the report	Content	~No. pages
Title	• Summarizing the project's vision and solution in one catchy sentence	
*It is up to you how you present the items below. It is important that you justify your choices/insights, reveal significant issues in relation to your vision, and describe insights in clear and understandable language. We impose the use of figures in specific sections whenever the word "*FIGURE" appears. *All the following elements should seek to incorporate relevant scientific literature – recommended during the classes, linked to your previous courses, found by yourself, or by others.*		
Introduction	• Present timeline of all the ideas since the initial brainstorm and problem identification and design choices (at least one FIGURE) ○ In weeks 1-2 write down what you expect to have achieved by the end of the course. Do not edit it. You will reflect on it in weeks 14-15 while preparing this report. • Brief overview of the context and the chosen problem.	1 - 3
Background, related work, and problem statement	• What is your vision of the world? *With vision, we mean a description of your project's overall goals and purpose. The vision should be positively engaging, clear, and strong, and indicate that you are passionate about the project. The intention is to encourage you to be clear and focused already at this stage, although, at later stages, you would have to be able to identify needs that you have not thought of yet, and welcome design ideas that may come as a surprise.* • What is the problem 'pain point' (problem formulation) of which stakeholder? • Why is it interesting and worth investigating? *(e.g., how many people suffer from it, how often? What is the 'cost' of not having the solution? 'cost' = time, money, emotional distress….)* • What are the possible competitors already in the market? (ideally in the form of a comparison TABLE)	1 - 3
User needs	• The main findings from your fieldwork so far, and a summary of the outcomes of the surveys/interviews conducted with potential users/stakeholders. *This includes information about their needs, challenges, wishes, the scope of practice, and important artifacts. You are encouraged to use various methods to outline and describe this (e.g., illustrations, photographs, diagrams, etc.).*	2 - 4
Methodology and design process (empirical production)	• What methodology did you use to investigate the problem you identified? *(data from…, individual interviews, focus groups, observations, etc.)* • Who have you talked to, and why them? *(collection/selection of participants)* • What were the defined use cases, personas, and scenarios? (at least one FIGURE) • What was the primary human factor and/or emotion you designed for? (e.g.,	2 - 4

	convenience, calm state of mind, reduction of cognitive load, curiosity, shame, fear, surprise, joy, hope, trust, motivation, belonging, desire, …) Note: the more challenging the human factor and/or emotion, the wackier it gets!	
Results (the design(s))	• Which requirements have you defined for the solution you designed? Which requirement has been imposed by which user? (ideally in a TABLE) ○ functional requirements ○ non-functional requirements • Describe and visualize your design process. *(sketches, pictures)* (at least one FIGURE) • Present the prototype(s), use-case diagrams/interaction design, and use context. (at least one FIGURE) • Present the high-level architectural design (e.g., client-server (s) architecture) theoretically. (at least one FIGURE) Note: *We must also have access to any interaction design prototypes.* • Evaluation: How have you chosen to evaluate it? What are the results? What did you find out? How did you learn? What did you change? • Note: On the exam-pitch day, we will conduct a random test of your interaction design prototype.	4 - 6
Results ***Mobile Systems and Services*** (the implementation)	• Present the high-level operationalized software architectural design and implementation details Note: The system must integrate at least one publicly available, open API that provides dynamic (real-time or frequently updated) data. The API integration must be fully functional and demonstrated live during the exam-pitch day. • List the system's main software components, including those developed by you and any significant open-source components integrated into the project. ○ For each component, present shortly and visualize its primary function, inputs, outputs, and how it interacts with other components, and visualize different data flows (at least one FIGURE) ○ API Specification (at least one FIGURE, e.g., request-response sequence diagram) ○ API purpose (internal vs external, REST/GraphQL, etc.) ○ Base URL and versioning ○ Authentication method ○ Main endpoints (e.g., Endpoint URL, HTTP method, request parameters, request body schema, response schema, status codes) ○ Example request/response (e.g., JSON) ○ Error handling model ○ If applicable: rate limits, security considerations ○ *Note: The application must gracefully handle API failures, but the core functionality must depend on the live API.* ○ Present a user profile model (e.g., relational model, UML, ER, JSON schema) (at least one FIGURE) ○ Provide a pseudocode for the core algorithm(s) of your service (at least one FIGURE) ○ Present a model of a database(s) you created (e.g., relational model, UML, ER, JSON schema) (at least one FIGURE) ○ Specify the technologies, programming languages, and frameworks used for each component ○ Detail the testing strategies employed, the types of tests conducted (unit, coverage, etc.), and the summary of the outcomes of these tests. • Note: We must also have access to any interactive prototypes.	8 - 16

	• Note: You *may annex any additional significant code snippets or algorithms developed by you. Additionally, you may annex any supplementary materials that offer deeper insights into the architecture, the API, the service components, or their testing.* • Note: On the exam pitch day, we will conduct a random test of your app	
Discussion	• What did you learn about the design process and methodology - the main insights? • What worked and what didn't? • How would you approach it differently if you had to do it again? • What are the 3 top ethical design challenges you have encountered, and how have you addressed them? • Did you achieve what you imagined having by the end of the course? • (if applicable) Is your contribution a medical device (and why?) - discuss the validation requirements.	2 - 4
Conclusive Remarks	• What is your contribution to the current state of the art? *(short repetition)* • Is your contribution answering the problem ('pain point')? Of which stakeholder(s)? • What does the future look like? Where is more research needed to create your vision?	2 - 3

ANNEXES

Obligatory Annexes

- If 'Mobile Systems and Services': Documentation pages from the API you used
- Your final pitch presentation slide deck

Optional Annexes

- Brainstorm outputs, 10+10.. and any other tangible (e.g., can be written by hand, scanned) inputs from earlier parts of the course
- Your solution's business model canvas
- Your solution's ethics canvas
- Your solution's privacy policy
- …